

TEST YOUR UNDERSTANDING

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Levels 1-2: Remembering/Understanding

1. Fat digestion yields fatty acids and glycerol. Protein digestion yields amino acids. Both digestive processes
(A) occur inside cells in most animals.
(B) add a water molecule to break bonds.
(C) require a low pH resulting from HCl production.
(D) consume ATP.
2. The mammalian trachea and esophagus both connect to the
(A) pharynx.
(B) stomach.
(C) large intestine.
(D) rectum.
3. In which organ does almost all enzymatic digestion of food occur?
(A) stomach
(B) small intestine
(C) large intestine
(D) pancreas
4. In which digestive system organ does nearly all nutrient absorption occur?
(A) stomach
(B) small intestine
(C) large intestine
(D) pancreas

Levels 3-4: Applying/Analyzing

5. If you put the following events in the order they occur in the human digestive system, which would be the third event in the series?
(A) Cells in gastric pits secrete protons.
(B) Pepsin activates pepsinogen.
(C) HCl activates pepsinogen.
(D) Partially digested food enters the small intestine.
6. After surgical removal of the gallbladder, a person might need to limit his or her dietary intake of
(A) starch. (C) sugar.
(B) protein. (D) fat.
7. If you were to jog 1 km a few hours after lunch, which stored fuel would you probably tap?
(A) muscle proteins
(B) muscle and liver glycogen
(C) fat in the liver
(D) fat in adipose tissue

Levels 5-6: Evaluating/Creating

8. **DRAW IT** Create a flowchart to summarize the events that occur after partially digested food leaves the stomach in a human. Use the following terms: bicarbonate secretion, circulation, decrease in acidity, increase in acidity, secretin secretion, signal detection. Next to each term, indicate the compartment(s) involved. You may use terms more than once.

9. **EVOLUTION CONNECTION** Lizards and snakes cannot breathe while chewing food because the connection between their external nostrils and their esophagus is in the mouth. In contrast, mammals can continue breathing through the nostrils while chewing food in the mouth. However, choking sometimes occurs when the paths of air and food cross each other. Thinking about the high oxygen demand of active endotherms, explain how the concept of descent with modification explains this “imperfect” anatomy of some amniotes (see Figure 34.2 to review vertebrate phylogeny).
10. **SCIENTIFIC INQUIRY** In human populations of northern European origin, the disorder called hemochromatosis causes excess iron uptake from food and affects one in 200 adults. Among individuals 15 to 50 years old, men are ten times as likely as women to suffer from iron overload. Taking into account this range of ages, propose a hypothesis that explains this difference.
11. **WRITE ABOUT A THEME: ORGANIZATION** Hair is largely made up of the protein keratin. In a short essay (100–150 words), explain why a shampoo containing protein cannot replace the protein in damaged hair.
12. **SYNTHESIZE YOUR KNOWLEDGE**



Owls, which have a varied diet, periodically cough up pellets composed of indigestible material, such as bones, feathers, and fur. (a) Compare this aspect of the owl's digestive process to those of the hydra, the rabbit, and the cow. (b) In which digestive compartment do you predict owl pellets form (see Figure 41.7c)? Explain.

For selected answers, see Appendix A.

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➔ **Instructors:** Questions can be assigned in Mastering Biology.